

Distributional Empirics And Scope Of The Paper

Date: 2026-05-08

This note clarifies how the paper should use cross-sectional facts about fertility over the income, wealth, housing-space, tenure, and location distributions. These facts do not “validate” the model in a causal sense. They are equilibrium objects. Their role is to discipline the mechanism and define what the current paper can and cannot claim.

1. The Right Interpretation

The paper studies a housing-space mechanism: children raise the demand for space, and this interacts with tenure, wealth, and location. That is not a full theory of fertility over the income distribution. The fertility literature makes this distinction important. Becker–Lewis highlights quantity-quality and child costs. Jones, Schoonbroodt, and Tertilt show that standard models struggle to explain the income-fertility relationship without extra structure. Doepke, Hannusch, Kindermann, and Tertilt emphasize that the old negative income-fertility facts have weakened or reversed in parts of the high-income world. Bar, Hazan, Leukhina, Weiss, and Zoabi show why high-income fertility can rise when households can marketize childcare and home production.

So the paper should not claim: “housing costs explain fertility by income.” That would be too strong. The defensible claim is narrower:

Cross-sectional gradients in income, wealth, tenure, space, and location are used as distributional discipline. A credible housing-space model must match the birth-related housing response while remaining consistent with where fertility occurs in the joint distribution of resources and housing.

This says the model must not fix one targeted margin by breaking the larger empirical distribution.

2. What We Know Empirically

The existing empirical package already supports five mechanism-adjacent facts.

First, PSID first-birth and second-birth event studies show that rooms rise after births. These are the cleanest direct facts for the housing-space block. They discipline child-related housing requirements, not a sequential fertility hazard.

Second, birth changes the composition of moves. The broad “any move” margin mixes lifecycle mobility, anticipation, and post-birth lock-in. The clean fact is narrower: moves explicitly for more space rise around first birth.

Third, families are spatially sorted. The ACS/MMS center-periphery build shows that parents are less central than non-parents, and movers with children sort more toward peripheral destinations within metros. This is the right spatial equilibrium fact, but it is not a PSID event study of birth-induced flight to cheaper locations.

Fourth, ownership is central. The PSID wealth/fertility work shows a large within-wealth ownership-fertility gap among young renters. This is closer to the model than a simple wealth gradient because the relevant object is wealth as it affects tenure and family-sized housing.

Fifth, the physical room distribution matters. Matching birth-related room increments is not enough. The model must also put childless renters, childless owners, and families in plausible parts of the room distribution.

3. How To Speak About The Main Distributions

Income

Income is the most dangerous object. The live model has common wages across center and periphery and no persistent earnings type. It can speak only indirectly to income gradients, through age profiles, saving, tenure, and sorting. If the data show an income gradient conditional on wealth, tenure, age, and location, the current model is missing an earnings/type or time-cost channel. Income gradients should therefore be descriptive context and possible external discipline, not a main validation target.

Wealth

Wealth is closer to the model, but it must be decomposed. Liquid wealth, total wealth, housing wealth, and ownership are different objects. Existing PSID outputs do not support “richer households have more children.” They point toward a hump-shaped parenthood profile over young-adult wealth: low wealth appears constrained, middle and upper-middle wealth look most fertile, and the top is lower or delayed. The model should therefore report fertility by liquid-wealth bins, tenure, and location. If it generates monotone fertility increasing in wealth because Stone-Geary needs bind only the poor, that is a failure unless the empirical bins also say so.

Space

Space is central. The model should be judged on both dynamic housing responses and level distributions. PSID disciplines room changes around births. ACS disciplines where households sit in levels. The model should match childless renter rooms, childless owner rooms, parent rooms, and the mass on lower and middle housing rungs before family-space policy counterfactuals are treated as informative.

Tenure And Housing Wealth

The housing-fertility literature is clear that owner and renter effects differ. Lovenheim and Mumford and Dettling and Kearney show why higher house prices can raise fertility for owners through wealth but lower it for non-owners through costs. The model should therefore avoid one generic “housing-cost effect.” The relevant split is:

- renters and would-be buyers face higher child-related housing costs;
- owners also hold housing wealth;
- liquid constraints and down payments matter mainly around tenure transitions;
- ownership may be a family-space technology, not only a wealth proxy.

4. What The Current Paper Can Claim

The current paper can claim that housing space is a quantitatively disciplined channel linking fertility, tenure, and spatial sorting. The same model must confront birth-related room increases, moves for space, parental sorting toward less central places, ownership/fertility gradients, and the physical distribution of rooms.

The paper can also claim that cross-sectional gradients are not nuisance facts. They are acceptance tests. A benchmark is not policy-ready if it matches TFR and birth-room increments but puts childless renters in implausibly large units or generates the wrong fertility profile by wealth and tenure.

The paper can also use the literature to motivate why the answer is not “more resources always imply more fertility.” Budget constraints, housing constraints, ownership wealth, time costs, education, career

costs, bargaining, and childcare marketization can all matter. This paper isolates the housing-space force and embeds it in spatial equilibrium.

5. What The Current Paper Cannot Claim Yet

The paper cannot currently claim to explain the entire fertility-income relationship; it does not have the state space for that. It cannot claim a direct PSID event study from first birth to cheaper MSA or cheaper local housing market. It cannot claim lower parent price elasticity unless a cleaned price panel is built. It cannot claim that low-wealth parents are the ones making spatial flight moves. And it cannot claim that Stone-Geary non-homotheticity alone is the final mechanism. If the model fails the distributional checks, the natural additions are persistent income heterogeneity, time costs, childcare marketization, or fertility-taste heterogeneity.

6. The Practical Empirical Object

The right deliverable is one distributional discipline table with data and model analogs:

Object	Data role	Model role
Parenthood / completed fertility by income bins	descriptive boundary	checks whether missing income heterogeneity matters
Parenthood / completed fertility by liquid-wealth bins	constraint evidence	checks whether fertility is too monotone in assets
Fertility by ownership and housing wealth	owner/renter split	checks tenure and housing-wealth channel
Rooms by parent status and tenure	space levels	disciplines housing menu and baseline needs
Room changes around births	dynamic space demand	disciplines child-related housing needs
Center/periphery by parent status	spatial sorting	disciplines equilibrium location choices
Moves for space around birth	mechanism support	checks whether birth induces housing adjustment

This table should be described as a discipline table, not as a causal validation table. That distinction is the clean way to position the empirics.

References To Keep In View

- Becker (1960); Becker and Lewis (1973): quantity-quality and child-cost foundations.
- Jones, Schoonbroodt, and Tertilt (2008/2011): standard theories struggle with the fertility-income relationship.
- Doepke, Hannusch, Kindermann, and Tertilt (2022): the high-income-country fertility facts have changed; old negative gradients are not universal.
- Bar, Hazan, Leukhina, Weiss, and Zoabi (2018): high-income fertility and marketization of childcare/home production.
- Lovenheim and Mumford (2013); Dettling and Kearney (2014): housing wealth and house-price effects differ sharply by tenure.
- Kulu (2013): housing conditions matter especially for first-birth spatial fertility differences.

- Albouy, Ehrlich, and Liu (2016); Davis and Ortalo-Magne (2011): housing demand and expenditure-share facts discipline the housing block.